

Week 1 – Task 1: AI Tool Exploration & Report

Instructions:

- Use ChatGPT to explain AI in simple words.

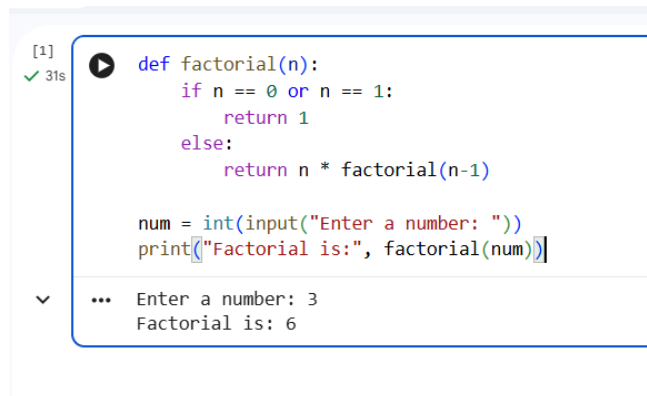
Explanation:

Artificial Intelligence (AI) is a technology that enables machines to think, learn, and make decisions like humans. It allows computers to perform tasks such as recognizing images, understanding language, solving problems, and making predictions.

AI works by using data and algorithms to learn patterns and improve performance over time.

- Generate Python code using ChatGPT.

Prompt Used: Write a Python program to calculate the factorial of a number.

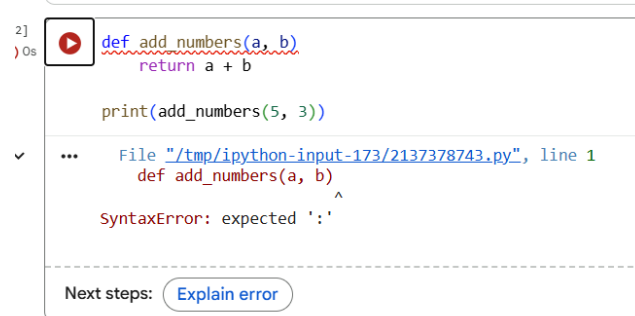


```
[1] ✓ 31s def factorial(n):  
    if n == 0 or n == 1:  
        return 1  
    else:  
        return n * factorial(n-1)  
  
    num = int(input("Enter a number: "))  
    print("Factorial is:", factorial(num))  
  
... Enter a number: 3  
Factorial is: 6
```

Explanation:

This program defines a function `factorial()` that uses recursion to calculate the factorial of a given number.

- Create a small Python error and ask ChatGPT to debug it.



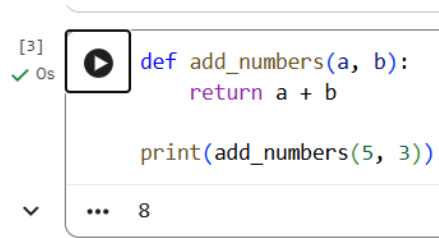
```
2] ) 0s def add_numbers(a, b)  
    return a + b  
  
print(add_numbers(5, 3))  
  
... File "/tmp/ipython-input-173/2137378743.py", line 1  
    def add_numbers(a, b)  
                        ^  
SyntaxError: expected ':'  
  
Next steps: Explain error
```

Prompt Used:

“Why is this Python code giving an error?”

AI Explanation:

The error occurs because a colon (:) is missing after the function definition.



```
[3]
✓ 0s def add_numbers(a, b):
      return a + b

      print(add_numbers(5, 3))
... 8
```

- **Write what you learned from this activity.**

- AI tools like ChatGPT can simplify complex concepts.
- AI can generate working Python code quickly.
- AI helps in debugging errors efficiently.
- It improves productivity and saves time.
- AI can assist beginners in learning programming step by step.

This activity helped me understand how AI tools automate explanation, coding, and troubleshooting tasks.

- **Mention one limitation of AI tools.**

One limitation of AI tools is that they can sometimes generate incorrect or incomplete answers. Therefore, we must have to verify the information and not rely blindly on AI-generated outputs.